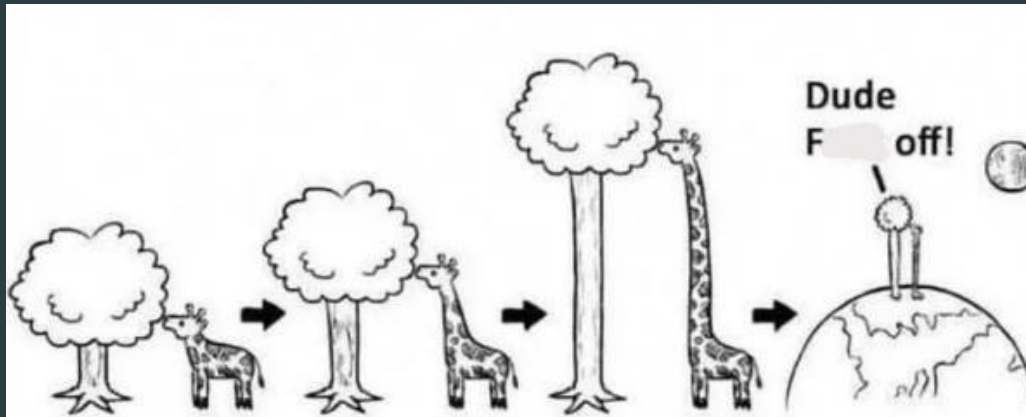




UUV Genetic Algorithm Optimization

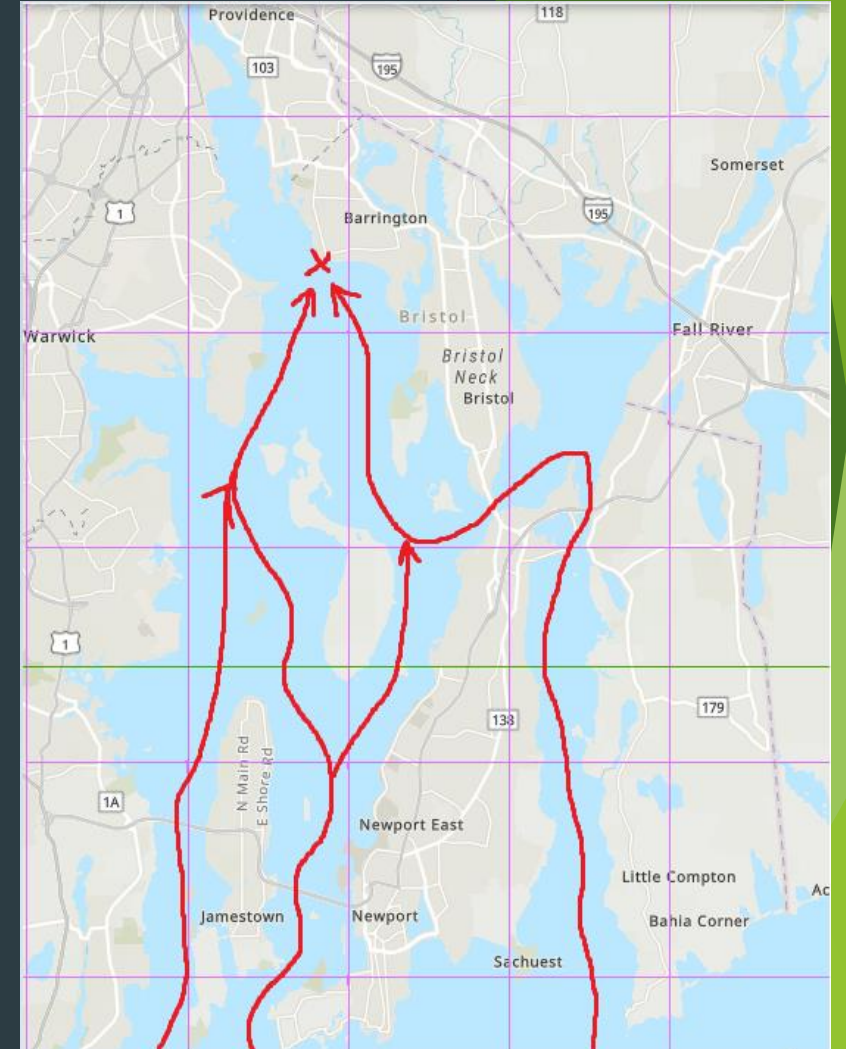
Gunner Cook-Dumas

Why use Genetic Algorithm

- ▶ Explore multiple best options
- ▶ Don't need to explicitly program it
- ▶ Can generate multiple optimal results
- ▶ Can use the results to optimize other AI frameworks

Problem

- ▶ Attacking UUV choose an optimal path towards target
- ▶ Optimize the placement of detectors
- ▶ See which scenario detects the most and cost the least
- ▶ Get the best scenario



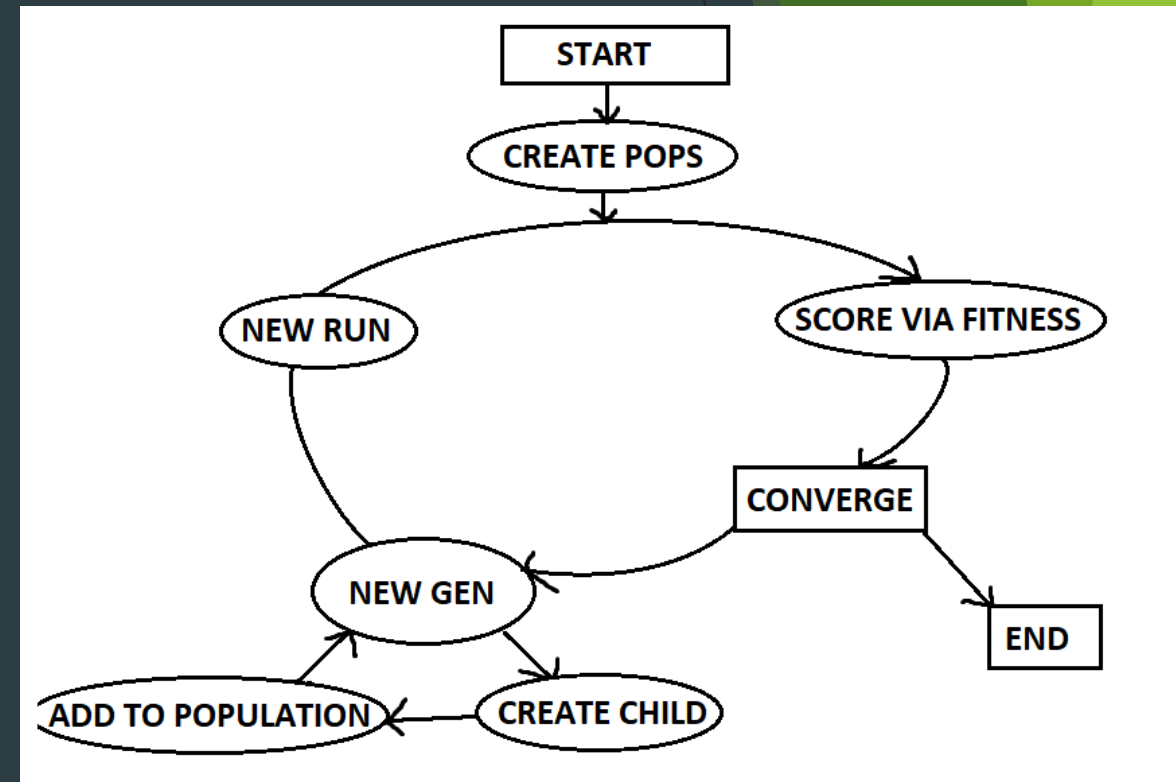
What have I done

- ▶ Model keeps a population scenario
- ▶ Each scenario is a chromosome
- ▶ Chromosomes contain all the data and objects
- ▶ Viable spawn locations chosen by user
- ▶ The genetic algorithm iterates through generations towards convergence



Overview of the Genetic Algorithm (GA)

- ▶ Contain both GA population and agent population
- ▶ Agent population is static
- ▶ GA population converges towards goal
- ▶ every run determine top 2 GA via fitness score
- ▶ Remove all agents in the populations
- ▶ Refill the GA population via mating



The Chromosome

- ▶ Each chromosome acts as a different scenario
- ▶ # of detectors is the max amount
- ▶ List of agents contains the agent objects
 - ▶ Position
 - ▶ Detected attacker
 - ▶ Destroyed attacker
- ▶ The total cost of the chromosome

Chromosome

# of detectors	list of agents	total cost
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list of agents = {(agent one), (agent two), ..., (agent n)}

Fitness

- ▶ Fitness determines how well the chromosome did
- ▶ Alpha and Beta are weights
- ▶ $Fitness = \alpha \left(\frac{attackers\ detected}{total\ \#\ of\ attackers} \right) - \beta(cost)$

Mating

- ▶ The top two are chosen for mating A and B
- ▶ Randomly select number of detectors
- ▶ Randomly select agent list from A or B
 - ▶ Choose agent randomly from said list
 - ▶ Loop back
- ▶ Change to mutate to introduce new genes
 - ▶ Simple subtract or add a new agent
- ▶ Total cost = number of detectors * cost

$A = \{5, (1A, 2A, 3A, 4A, 5A), c\}$

$B = \{3, (1B, 2B, 3B), c\}$

$child = \{3, (1B, 3A, 3B), 150\}$

Plan to do

- ▶ Update the fitness function
- ▶ Prioritize successful agent selection during mating
- ▶ Find good mutation rate
- ▶ Allow user to specify settings
- ▶ Update all agents
- ▶ Fix pathfinding